

Merlin Yang

Oakland

yang.mengy@northeastern.edu | (831) 251-4873 | github.com/merlinymy | merlinyang.com

SUMMARY

I build digital experiences where design, performance, and engineering meet. I care obsessively about polish, responsiveness, and user delight. From fine-grained animations to clean, scalable architecture, I'm looking for a role where I can push the boundaries of front-end engineering and ship products that feel as good as they look.

SKILLS

React 19, TypeScript, JavaScript, Node.js, Python, Three.js, Tailwind, Express, Supabase, PostgreSQL, Docker, Vite, Git, CI/CD, WebSocket

PROJECTS

FireProofSheep – Top-Down Arcade Style Web Game

Sep 2025 - Present

Game Developer

- Built a real-time multiplayer architecture using Socket.IO with authoritative server logic, delta-compressed state sync, and cooperative sessions—experience directly relatable to distributed experiment infrastructure.
- Designed systems for procedural generation, resource simulation, and deterministic state updates across clients and servers.
- Optimized collision + serialization logic, reducing computation 85%, network traffic 75%, and maintaining 1–2 ms server tick times—strong foundation for high-throughput, low-latency pipelines.
- Maintained testing discipline (Vitest), internal documentation, and continuous performance profiling—mirroring reliability expectations of production infra roles.

AI-Assisted Social Media Post Generator

Jul 2025 - Aug 2025

Full-Stack Developer

- Built an end-to-end Supabase-authenticated React SPA with protected routing, multilingual UI, and fast iteration cycles.
- Designed and implemented a full content-generation workflow: AI text/image pipelines, editable previews, and modal-based publishing.
- Integrated OpenAI Responses API for on-device AI workflows, storing outputs securely in Supabase—experience bridging research-style prototyping with production-ready code paths.
- Shipped quickly with clean abstractions, modular components, and tight UI engineering.

TCG Memory League - Memory Game

Apr 2025 - Apr 2025

React Developer

- Rebuilt a classic memory game into a large-scale collectible system featuring 10,000+ cards, unlockable sets, and endless progression loops.
- Implemented performant UI systems using React 19 + Vite 6 with parallax/flip animations, adaptive audio, and localStorage persistence.
- Deployed to Vercel with instrumentation for performance and user behavior analytics—aligning with the evaluation/observability mindset of the role.

EDUCATION

Northeastern University

Sep 2025 - Jul 2027

Master's in Computer Science